

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester B.Tech (CBCS) Degree Examinations
 May 2024
MA1003-1 – DIFFERENTIAL EQUATIONS AND LAPLACE TRANSFORMS
 (ACT, EC, EE, VLSI)

Duration: 3 Hours

Max. Marks:100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

- The degree of the differential equation $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 7y = 0$ is _____
 A) 1 B) 2
 C) 3 D) 4
- The integrating factor of $\frac{dy}{dx} + y = e^{-x}$ is _____
 A) e^{-x} B) e^x
 C) x D) $-x$
- The orthogonal trajectories of the family of curves $xy = a$ is _____
 A) $x^2 - y^2 = c$ B) $x^2 + y^2 = c$
 C) $x^2y = c$ D) $xy^2 = c$
- The general solution of $y = px - e^p$ is
 A) $y = cx$ B) $y = cx + e^c$
 C) $y = cx - e^c$ D) $y = e^c$
- The particular integral of $(D + 3)^2y = 2$ is _____
 A) $\frac{2}{9}$ B) 2
 C) 1 D) 9
- $y = (C_1 + C_2x)e^x$ is the complete solution of which of the following differential equations
 A) $(D^2 - 1)y = 0$ B) $(D^2 + 2D + 1)y = 0$
 C) $(D^2 - 5)y = 0$ D) $(D^2 - 2D + 1)y = 0$
- The complete solution of $(D^2 - 4D + 4)y = 0$ is _____
 A) $y = (C_1 + C_2x)e^x$ B) $y = e^x(C_1 \cos \sqrt{3}x + C_2 \sin \sqrt{3}x)$
 C) $y = (C_1 + C_2x)e^{2x}$ D) $y = (C_1 + C_2x)e^{3x}$
- The linear differential equation with constant coefficients corresponding to the equation $x^2 \frac{d^2y}{dx^2} + 3x \frac{dy}{dx} + y = \frac{1}{(1-x)^2}$ is _____
 A) $(D^2 + 1)y = \frac{1}{(1-e^t)^2}$ B) $(D^2 + 3D)y = \frac{1}{(1-e^t)^2}$
 C) $(D^2 + 4D)y = \frac{1}{1-e^t}$ D) $(D + 1)^2y = \frac{1}{(1-e^t)^2}$
- $L\{\sin t\} =$ _____
 A) $\frac{1}{s^2 - 1}$ B) $\frac{1}{s^2 + 1}$
 C) $\frac{1}{s^2 + 2}$ D) $\frac{1}{s^2 - 3}$

10. $L\{\cos t\} =$ _____
 A) $\frac{1}{s^2 - 1}$
 C) $\frac{1}{s^2 + 1}$
 B) $\frac{1}{s^2 + 1}$
 D) $\frac{1}{s^2 - 1}$
11. $L\{10\} =$ _____
 A) $\frac{10}{s^2}$
 C) $\frac{10}{s^2 - 1}$
 B) $\frac{10}{s}$
 D) $\frac{1}{s^2 - 1}$
12. $L\{e^t \sin 2t\} =$ _____
 A) $\frac{-2}{(s-1)^2 + 4}$
 C) $\frac{1}{(s-1)^2 + 4}$
 B) $\frac{2}{(s-1)^2 + 4}$
 D) $\frac{4}{(s-1)^2 + 4}$
13. $L^{-1}\left\{\frac{1}{s}\right\} =$ _____
 A) 1
 C) 3
 B) 2
 D) 4
14. $L^{-1}\left\{\frac{1}{s^2 + 9}\right\} =$ _____
 A) $\frac{\sin 3t}{3}$
 C) t^2
 B) $\frac{\cos 3t}{3}$
 D) t
15. $L^{-1}\left\{\frac{6}{s^4}\right\} =$ _____
 A) t^2
 C) t
 B) 1
 D) t^3
16. If $L\{f(t)\} = \bar{f}(s)$ then $L\left\{\frac{f(t)}{t}\right\} =$ _____
 A) $s \int_s^\infty \bar{f}(s) ds$
 C) $\int_s^\infty s \bar{f}(s) ds$
 B) $\int_s^\infty \bar{f}(s) ds$
 D) $\int_s^\infty s^2 \bar{f}(s) ds$
17. The order of the partial differential equation $\frac{\partial^2 z}{\partial x^2} = 10xy + 7$ is _____
 A) 1
 C) 3
 B) 2
 D) 4
18. The partial differential equation obtained from $z = axy + b$ by elimination of arbitrary constants a and b is _____
 A) $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = 0$
 C) $x \frac{\partial z}{\partial x} = y \frac{\partial z}{\partial y}$
 B) $2x \frac{\partial z}{\partial x} = y \frac{\partial z}{\partial y}$
 D) $x \frac{\partial z}{\partial x} = 2y \frac{\partial z}{\partial y}$
19. The partial differential equation obtained from $z = f(x^2 - y^2)$ by elimination of arbitrary function is _____
 A) $x^2 \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = 0$
 C) $y \frac{\partial z}{\partial x} + x \frac{\partial z}{\partial y} = 0$
 B) $x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y} = 0$
 D) $y \frac{\partial z}{\partial x} - x^2 \frac{\partial z}{\partial y} = 0$
20. The solution of the partial differential equation $\frac{\partial^2 z}{\partial x^2} = xy$ is _____
 A) $z = \frac{x^3 y}{6} + xf(y) + g(y)$
 C) $z = \frac{x^3 y}{6} + f(x)$
 B) $z = \frac{x^3 y}{6} + xf(x) + g(y)$
 D) $z = \frac{x^3 y}{6} + g(y)$

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	Solve $\frac{dy}{dx} + \frac{y}{x} = x^2 y^6$.	6	L*2	1	1
b)	If a body which is originally at 80°C cools down to 60°C in 20 minutes, the temperature of air being 40°C. Find the temperature of the body after 50 minutes from the original.	6	L2	1	1
c)	Solve $(D^2 + 5D + 6)y = 2e^x + 3e^{2x}$.	4	L2	2	1
2. a)	Find the orthogonal trajectories of the family of curves $y = ke^{-2x} + 5x$.	6	L2	1	1
b)	Solve $y = 2px + \tan^{-1}(xp^2)$.	6	L2	1	1
c)	Solve $(D^2 + 2D + 1)y = 3 \sin x$	4	L2	2	1
3. a)	Solve $(D^2 + 1)y = \sec x \tan x$ using the method of variation of parameters.	6	L2	2	1
b)	Solve $\frac{d^2y}{dx^2} - \frac{2y}{x} = x + \frac{1}{x^2}$	6	L2	2	1
c)	Solve $y = px + \sin^{-1}(p)$.	4	L2	1	1
Unit – II					
4. a)	Find $L\{\int_0^t \frac{e^t \sin t}{t} dt\}$.	6	L2	3	1
b)	Find $L[t^2 e^{-3t} \sin 4t]$.	6	L2	3	1
c)	Find $L^{-1}\left\{\frac{4}{(s^2+4s+20)}\right\}$.	4	L2	4	1
5. a)	If $f(t)$ is a periodic function with period T, then prove that $\int_0^T e^{-st} f(t) dt$.	6	L2	3	1
b)	Rewrite $f(t) = \begin{cases} \sin t, & 0 < t < \frac{\pi}{2} \\ \cos t, & t \geq \frac{\pi}{2} \end{cases}$ using unit step function and find its Laplace transform.	6	L2	3	1
c)	Find $L^{-1}\left\{\frac{15}{(s^2+4s+13)}\right\}$	4	L2	4	1
6. a)	Using Convolution theorem find $L^{-1}\left\{\frac{1}{(s^2+a^2)^2}\right\}$	6	L2	4	1
b)	Solve $x''(t) + 4x'(t) + 4x(t) = 4e^{-2t}$; $x(0) = -1, x'(0) = 4$ by Laplace transform method.	6	L2	4	1
c)	Find $L\{5 \cos^2 4t\}$.	4	L2	3	1
Unit – III					
7. a)	Solve by the method of separation of variables $\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial t} + u$, where $u(x, 0) = 6e^{-3x}$.	6	L2	5	1
b)	Solve $(z - y)p + (x - z)q = y - x$.	6	L2	5	1
c)	Solve $(p^2 + q^2) = x + y$.	4	L2	5	1
8. a)	Derive one dimensional heat equation in the form $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$.	6	L2	5	1
b)	Form the partial differential equation by eliminating the arbitrary function from $x + y + z = f(x^2 + y^2 + z^2)$.	6	L2	5	1
c)	Solve by direct integration $\frac{\partial^2 u}{\partial x \partial y} = \cos(2x + y)$.	4	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
